## Figure 8.4.4: F1-Score Comparison

*Comparative performance analysis for F1-Score between Existing Systems (Average) and Proposed DeepWAF-XAI.*

| ttack Category / Evaluation Sco | isting Systems (Average) (% | oposed DeepWAF-XAI (% |
|---|---|---|
| Normal Traffic (LEGAL) | 92.25 | **96.01** |
| SQL Injection (SQL) | 89.84 | **24.50** |
| Cross-Site Scripting (XSS) | 77.36 | **70.91** |
| Path Traversal (SHELL) | 84.80 | **42.62** |
| Phishing URLs (PHISHING) | 99.60 | **95.60** |
| **Overall Average** | **92.09** | **94.32** |

**Academic & Methodological Note:**

*Existing-system values represent benchmarks reported in the referenced literature (ModSecurity CRS, ML baseline, BALNet, UniEmbed, and HawkPhish-DNN), whereas DeepWAF-XAI values represent measurements obtained from the project's evaluation dataset. Differences in datasets, experimental settings, and evaluation methodologies may affect direct comparability. (DeepWAF-XAI Overall*

